Electricity Generation

13.4

The large-scale production of electrical energy that we have today is possible because of electromagnetic induction. The electric generator, which provides electricity for most places in the world, relies on the law of electromagnetic induction to operate. An **electric generator** is a device that transforms other forms of energy into electrical energy. These other forms of energy can include thermal, gravitational, or kinetic energy. As you know from Chapter 5, the energy used to power generators can come from either renewable or non-renewable sources, each with its own benefits, disadvantages, and environmental impacts to be considered.

electric generator a device that transforms other forms of energy into electrical energy

In Section 13.3, you learned that alternating current was chosen for the transmission of electrical energy. So we will first look at the generation of alternating current.

The Alternating Current Generator

Electromagnetic induction requires a changing magnetic field to produce an electric current. In an AC generator there are two ways of changing the magnetic field. Either a permanent magnet can be spun inside a coil or a coiled conductor can be spun inside a magnetic field. We will examine a coiled conductor consisting of a single loop of wire rotating inside a magnetic field.

The AC generator shown in **Figure 1** shows a single loop of conducting wire set between the poles of two permanent magnets. There are two slip rings and two brushes. Each slip ring is connected to a different side of the loop. Slip ring 1 is connected to the left side of the loop, while slip ring 2 is connected to the right side of the loop. The slip rings rotate with the loop. The brushes are stationary and make contact with the slip rings to allow current to be directed out to the external circuit. The loop spins on the axis of rotation. The spinning force is provided by an external source of energy. For example, recall from Chapter 5 that at a hydroelectric power plant, the falling water turns the blades of a turbine connected to the electrical generator.

In Figure 1, the loop of the generator is being forced clockwise. Since the loop is moving inside the magnetic field provided by the external magnets, an electric current will be induced in the loop as described by Faraday's law. The direction of the induced current is predicted by Lenz's law. Remember that an induced current also creates an induced magnetic field around the loop. The induced magnetic field around the loop opposes the field that created it.

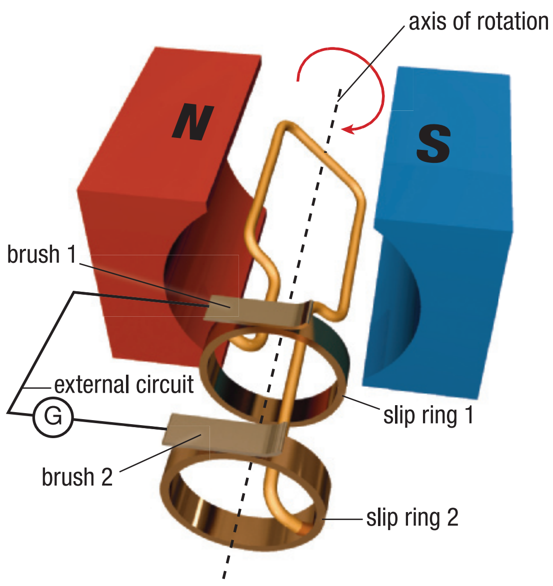

Figure 1 A single-loop AC generator

Let us examine the left part of the loop, in cross-section, closest to the north pole of the external magnet (**Figure 2**).

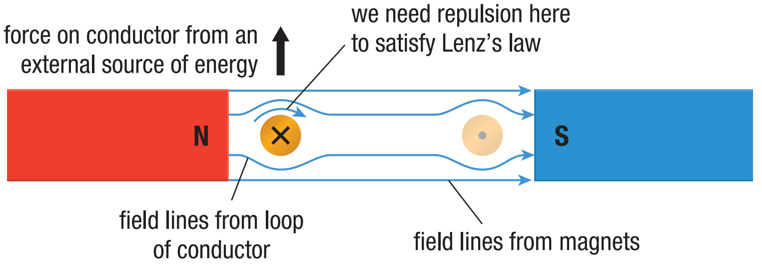

Figure 2 A cross-section view of the wire loop in the generator in Figure 1. An external source of energy moves the wire upward at the left side and downward at the right side.

The magnetic field around the wire opposes the external magnetic field. This happens because the field lines at the top of the wire are pointed in the same direction as the field lines from the external magnet. Magnetic field lines pointing in the same direction result in a force of repulsion. The force on the conductor coming from an external source of energy overcomes the magnetic repulsion force between the field from the conductor and the field from the magnets. Using the right-hand rule for a straight conductor, the conventional current points into the page.

Let us now examine the part of the loop, in cross-section, closest to the south pole of the external magnet (**Figure 3**).

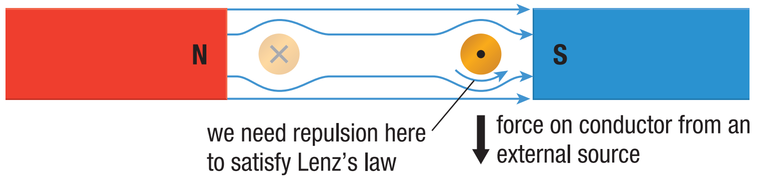

Figure 3 A cross-section view of the wire loop in the generator in Figure 1. An external source of energy moves the wire upward at the left side and downward at the right side.

On this side of the loop, the conductor is being forced downward, and the magnetic field from the loop opposes the external magnetic field. Again the field lines from the wire point in the same direction as the field lines from the external magnet and cause repulsion. Using the right-hand rule for a straight conductor, the conventional current points out of the page.

Therefore, the rotation of the loop in a clockwise direction in the magnetic field causes a conventional current in the loop in the direction shown in **Figure 4**. In the following Tutorial, we will investigate the direction of the current in the external circuit attached to the generator shown in Figure 4.

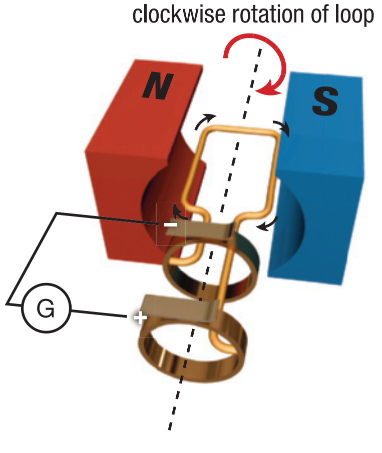

Figure 4 The direction of the induced current is shown by the black arrows around the loop. Note the positive and negative signs on the external circuit.

Tutorial 1 Explaining the AC Generator

As the loop of an AC generator spins, a current forms in the external circuit connected to the generator. What is the direction of the current in the external circuit?

In **Figure 5**, the current in the loop heads toward slip ring 2, which contacts brush 2, and into the external circuit as shown. Since the conventional current goes into the galvanometer at the positive terminal, the galvanometer needle points to the positive side of the scale.

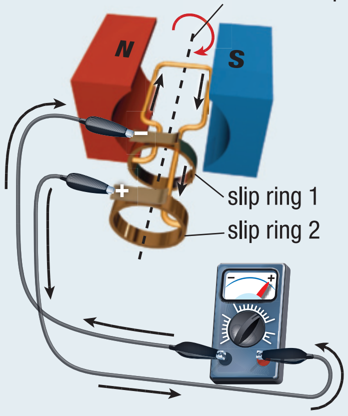

Figure 5

In addition to the factors discussed in Section 13.1, the amount of induced current also depends on the angle of the conductor in relation to the external magnetic field. The induced current is at a maximum when the plane of the loop is parallel to the external magnetic field. As the loop rotates toward 90° of rotation, the amount of current decreases. Once the loop is perpendicular to the magnets (or 90° of rotation), the current reads zero, as shown in **Figure 6**.

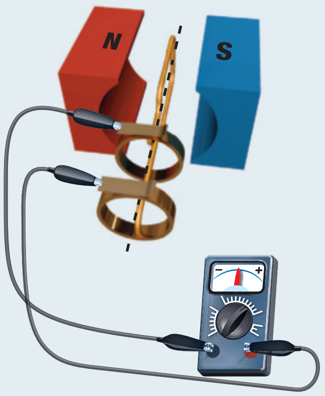

Figure 6

As the loop rotates from 90° of rotation and approaches 180°, the conventional current in the loop goes into slip ring 1. This reverses the direction of the current in the external circuit, as shown in **Figure 7**. Note that the conventional current goes into the galvanometer at the negative terminal. The galvanometer needle moves to the negative side.

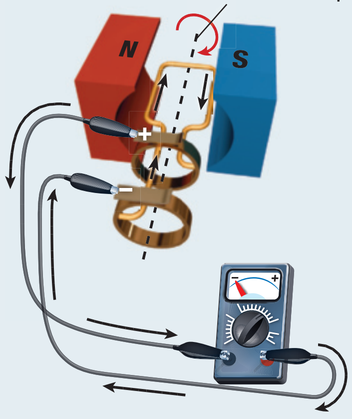

Figure 7

As the loop rotates away from 180°, the current once again decreases until it reaches zero at 270° of rotation, as shown in **Figure 8**.

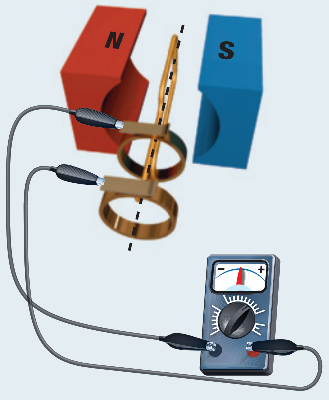

Figure 8

At this point, the current once again reverses direction and enters the external circuit at slip ring 2. This starts the whole process over again. The galvanometer readings are plotted on a graph in **Figure 9**. The rotation rate of the loop matches the frequency at which the current changes direction.

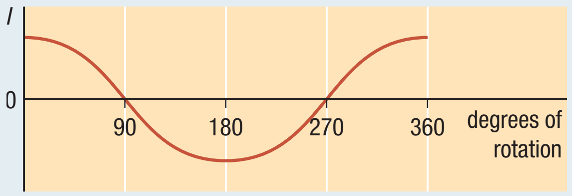

Figure 9

Mini Investigation

What Factors Affect Electricity Generation?

Skills: Planning, Performing, Observing, Communicating

SKILLS HANDBOOK A2.1, A2.4

Generators use coils and magnets to generate electricity. In this investigation, you will determine what factors affect the amount of current produced.

Equipment and Materials: galvanometer; 2 bar magnets; 2 coils with different number of windings; 2 alligator clip leads

1. Connect the coil with fewer windings to the galvanometer using the alligator clip leads.
2. Plan a procedure to test the factors that affect the amount of current produced.
3. Record your observations in a series of statements that are framed as follows: Changing the ___________ produced a maximum reading on the galvanometer of _______.

A. How did moving the magnet more quickly into the coil affect the amount of current? T/I

B. How did reversing the pole of the magnet inserted in the coil, while keeping the speed of the magnet going into the coil constant, affect the amount of current? T/I

C. How did using two magnets affect the amount of current? T/I

D. How did changing the number of windings in the coil affect the current? T/I

Factors Affecting Generator Output

The single-loop AC generator discussed in Tutorial 1 is useful for demonstration purposes. However, to increase the amount of current generated we could use a coiled conductor wrapped around a soft-iron armature. This increases the strength of the induced magnetic field. We could also rotate the soft-iron armature faster or use stronger external magnets. In Tutorial 2 we will look at using a coil in a generator.

Tutorial 2 Using a Coil in an AC Generator

A coil-type generator uses a coil wrapped around a soft-iron armature, as shown in **Figure 10**. This armature is rotated by an external source of energy. How does this design affect the direction of current in the external circuit?

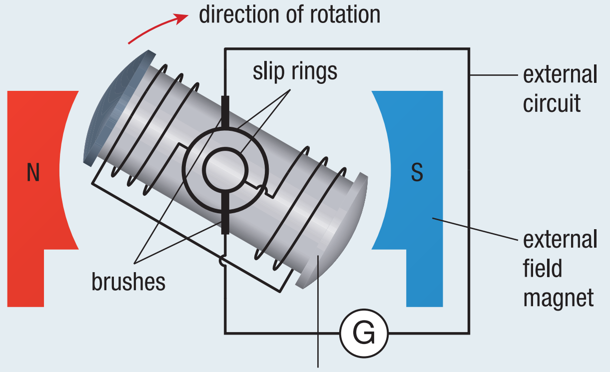

Figure 10

The rotation of the generator shown in Figure 10 is clockwise. As the shaded side of the armature moves away from the north pole of the external magnet, Lenz's law determines the left side of the armature to be a south magnetic pole. The armature is forced away from north. Thus, the induced magnetic field must oppose being pulled away by attracting with a south pole. Using the right-hand rule for a coil, the direction of the conventional current is as shown in **Figure 11**.

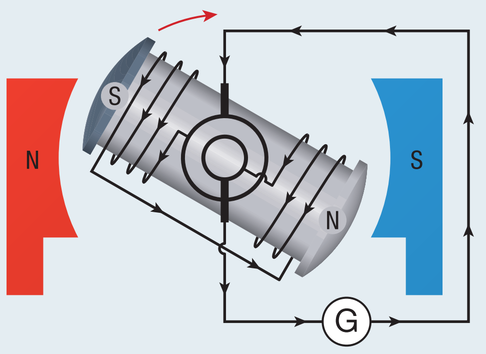

Figure 11

As the shaded side of the armature spins away from the north pole of the external field magnet, the amount of current increases to a maximum until the shaded side of the armature starts approaching the south pole of the external field magnet. The current increase can be explained by referring back to Tutorial 1. When the single loop was oriented so that it was perpendicular to the external magnetic field, the induced current was zero. When the single loop was oriented so that it was parallel to the external magnetic field, the induced current was at a maximum. The same applies whether you are using one loop or several loops. Using Lenz's law, the shaded side of the armature resists going toward south by repelling. This makes the shaded side of the armature a south pole. Using the right-hand rule, the direction of the conventional current is as shown in **Figure 12**.

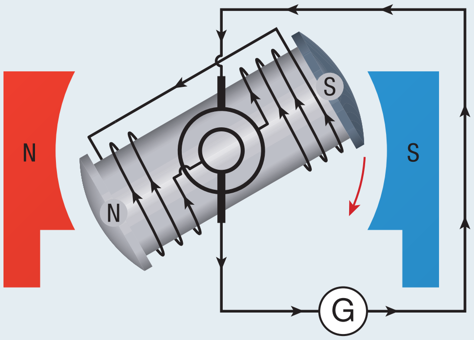

Figure 12

Now the shaded side of the armature spins away from the south pole of the external field magnet. This time Lenz's law predicts the shaded side of the armature to be north. This is because the shaded side of the armature is moving away from the south pole of the external field magnet. The shaded side of the armature must resist moving away by attracting to a north pole. The current now reverses direction as shown in **Figure 13**.

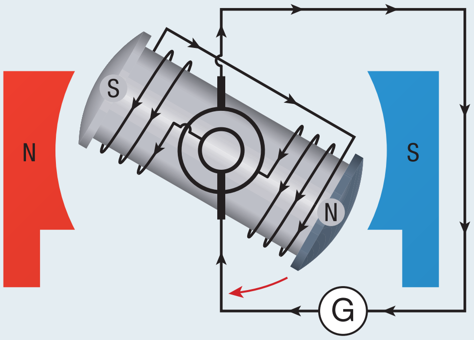

Figure 13

As the shaded side of the armature moves away from the south pole of the external field magnet, the current increases to a maximum value until the shaded side of the armature starts approaching the north pole of the external field magnet. The shaded side of the armature remains north on the side approaching north, causing repulsion. The current goes in the direction shown in **Figure 14**.

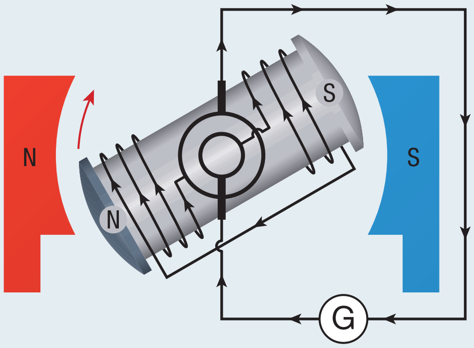

Figure 14

DC Generators

A DC generator, as shown in **Figure 15**, has the same design as a DC motor. It has a split ring commutator instead of slip rings. The split ring commutator serves to prevent the current from changing direction in the external circuit as it does in the AC generator. However, the induced current in the coil in the armature is still the same, as has been shown in the tutorials.

In the case of the DC motor, electrical energy is transferred into the motor to cause rotation or kinetic energy. In the case of a DC generator, rotation or kinetic energy (for example, from falling water, wind, or high-pressure steam) is used to turn the coil to generate electrical energy. Therefore, a generator may be considered a motor in reverse.

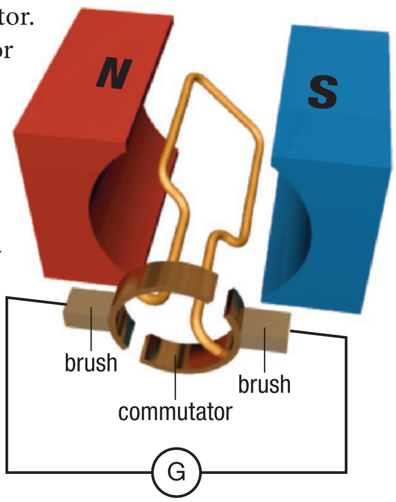

Figure 15 DC generator

Research This

Wind Turbines

Skills: Researching, Communicating, Identifying Alternatives

SKILLS HANDBOOK A5.1

Wind is a renewable resource, and many different designs of wind turbines are being engineered. Some are large and able to deliver enough electrical energy to power 5000 homes. Others are smaller and can power only one home. As you learned in this section, in a generator, the rotation rate of the loop is directly related to the frequency of the alternating current. Canada's electrical grid requires a frequency of 60 Hz. Different design philosophies are put in place to achieve this.

1. Choose a wind turbine technology. You can consider large-scale land-based turbines, off-shore turbines, or some of the novel small-scale turbines for residential applications.
2. Research how the technology works. Highlight the type of turbine used, the generator used, the type of technology used to control the electricity to feed it into the grid, and where the electricity is being used or developed.

A. At what rotation rate does the wind turbine spin, and how is the electricity controlled? T/I A

B. What are the maintenance considerations? K/U T/I

C. Prepare a tri-fold pamphlet highlighting your research. T/I C A

GO TO NELSON SCIENCE

UNIT TASK BOOKMARK

You can apply what you have learned about electric generators to the Unit Task on page 622.

13.4 Summary

- An electric generator transforms other forms of energy into electrical energy.
- Alternating current generators are designed with loops of a conductor that spin in a magnetic field. The ends of the loops are connected to two different slip rings, allowing them to produce alternating current.
- Coiling the conductor around an armature increases the strength of the induced magnetic field, making the generator produce more current.
- Spinning the armature more quickly or using a stronger external magnetic field also increases the current produced by the generator.
- DC generators are designed like a DC motor except energy is put into spinning the coil to generate electricity rather than putting electrical energy into the motor to cause it to spin.

13.4 Questions

1. Sketch the generator shown in Figure 1 on page 599, but change the rotation to counterclockwise. Answer the following questions based on your diagram. Consider the starting angle to be 0°. T/I C
   (a) At what angle(s) relative to your starting point would you expect maximum current in the loop?
   (b) At what angle(s) relative to your starting point would you expect a zero current?
   (c) Sketch a graph of the induced current in the external circuit.
   (d) What effect would reversing the polarity of the external magnets have on the current?
2. How does the rotation rate of the loop in the generator in Figure 1 compare to the frequency of the alternating current? K/U
3. How many times per second does a generator armature rotate in North America? K/U
4. The generator in **Figure 16** is rotated counterclockwise. Determine the polarity of the magnetic field on the armature, the direction of induced current in the coil, and the direction of current in the external circuit. T/I

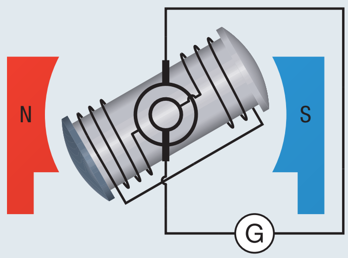

Figure 16